3.1 Importing data: vectors and rasters

QGIS lets you import vector and raster files, whether they are stored on your own computer or on a server.

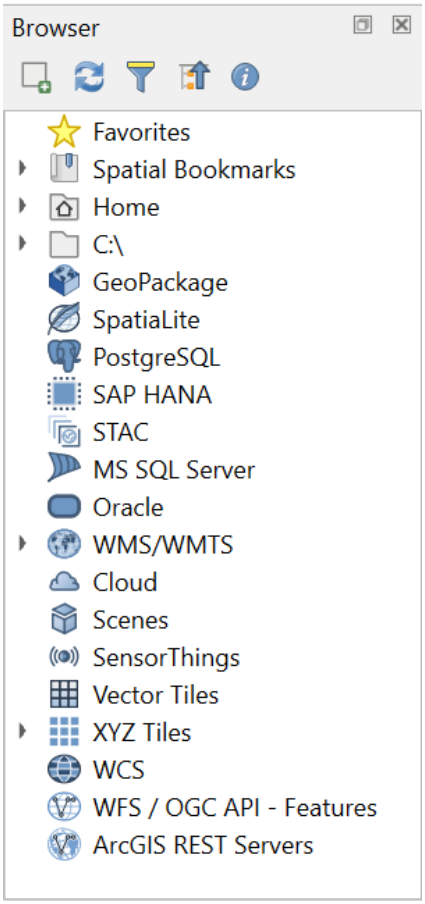
> Warning
>
> It is important to know that QGIS does no more than connect to the data source. **So the data themselves are never stored in the QGIS project**. If the data source is deleted, moved or renamed, the information from this source will no longer be available in your QGIS project (see The QGIS project).

Method 1: drag and drop

The simplest way to load geodata is to drag the files onto the QGIS interface: pick up the files in a folder on your computer and drop them on the map canvas or the *Layers* panel. For a shapefile, which consists of several files, you only need to select **the one with the extension .shp**.

> Tip
>
> *Drag and drop* is quick, but it **skips the import window** with its extra settings. When you need more control over how a file is imported, use the *Data Source Manager* instead, as you will see later in this learning path.

If you do not want to switch between windows all the time while working in QGIS, you can use the ***Browser* panel** instead: it gives you easy access to all the folders and files on your computer.



The *Browser* panel has another advantage: it can also hold **connections** to databases and servers. In this way, you have access not only to your local files but also to all kinds of other data sources, gathered in one handy window.

Tip

GeoPackages. You can load a GeoPackage just like any other file. If it contains one layer, that layer is added immediately; if it contains several layers, QGIS first opens a window in which you select the layers you want to add.

To make working with GeoPackages easier, you do not always have to load them as a whole. Instead, you can make a **connection** to a GeoPackage: in the *Browser* panel, right-click *GeoPackage* ▸ *New Connection...* and choose the `.gpkg` file. When you unfold the GeoPackage in the *Browser* panel, you see all its layers and can add exactly the ones you need. QGIS remembers this connection, also in other projects, so you only need to create such a connection once.

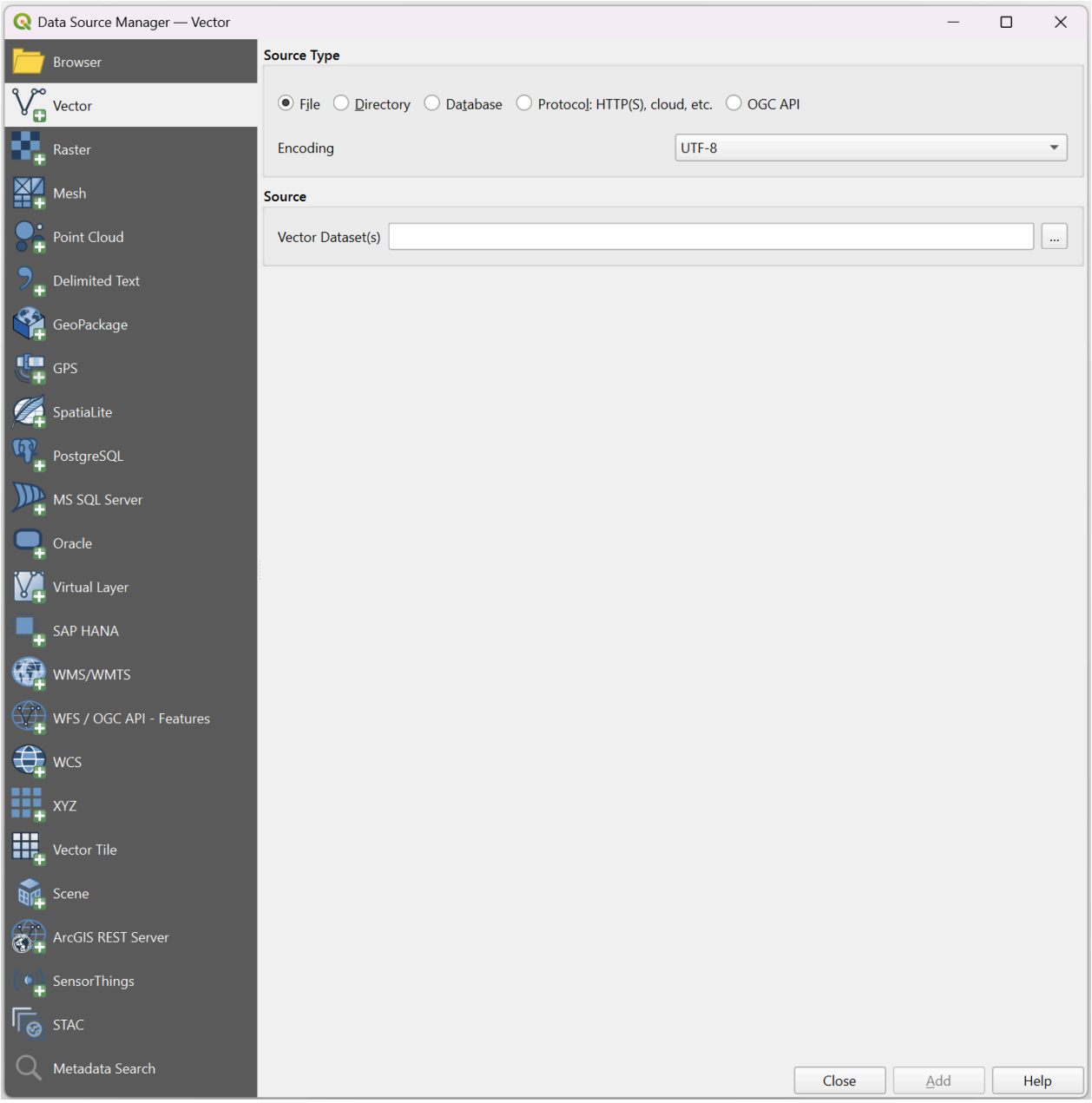
Method 2: via the Data Source Manager

You can also import data into QGIS through the *Data Source Manager*. Thanks to its many configuration options, this manager offers a more controlled way to load geo-information. You can open the *Data Source Manager* by choosing *Layer* ▸ *Data Source Manager* in the menu. Alternatively, you can click the first (leftmost) button of the *Data Source Manager Toolbar*, if this toolbar is shown (see QGIS interface: *View* ▸ *Toolbars*).

In the *Data Source Manager*, the navigation column on the left lists all the data source types you can load through the manager. If you click one of these types, the configuration options for this type appear on the

right. For some types, such as vector and raster files, these options are limited; for others you will have to configure more. We will give concrete examples of this in a later chapter, *Data in GIS (part 2)*: see for example Delimited text files (CSV).

The data source types you find on the left side of the *Data Source Manager* are also available when you choose *Layer* ▸ *Add Layer* in the main menu. Each of these submenu items takes you directly to the right tab of the *Data Source Manager*.



Vector data

To load locally stored vector data, choose the *Vector* tab and make sure that *Source Type* is set to *File*. Then select your file(s) with the … button next to *Vector Dataset(s)* and click *Add*.

Tip

A GeoPackage is a database, but one that is stored as a file on your computer: load it with *Source Type* ▸ *File*. The option *Database* is meant for databases on a server.

Raster data

In the same way, you can import local raster data through the *Raster* tab of the *Data Source Manager*. Vector and raster data are thus loaded in almost the same way, but through a different tab.

Make sure you always select the right tab. Otherwise QGIS gives an error message and you cannot import the data you want.